Hemodynamic Disorders
LG 7, Shock

Ass. Prof. Dr. Tenya Tariq Abdulhameed

M.B.Ch.B, M.Sc, Ph.D (HISTOPATHOLOGY)

College of Medicine, HMU.

2026-2027

Outline:

- Definition of shock
- Types of shock
- Pathogenesis of Septic Shock
- Stages of Shock
- Morphology
- Prognosis
- Summary

SHOCK



- Shock is a state in which diminished cardiac output or reduced effective circulating blood volume impairs tissue perfusion and leads to cellular hypoxia.
- At the outset, the cellular injury is reversible; however, prolonged shock eventually leads to irreversible tissue injury and is often fatal.

Types of shock:



1. **Cardiogenic shock:**

- results from low cardiac output as a result of myocardial pump failure.
- It may be caused by:
- myocardial damage (infarction),
- ventricular arrhythmias,
- extrinsic compression (cardiac tamponade) or
- outflow obstruction (e.g., pulmonary embolism).



2. Hypovolemic shock:

results from low cardiac output due to loss of blood or plasma volume (e.g., resulting from hemorrhage or fluid loss from severe burns).



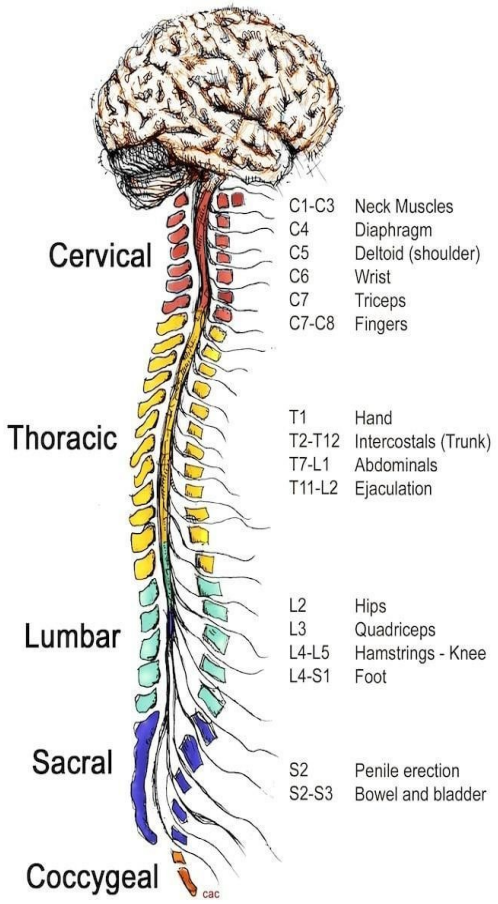
3. Septic shock:

- is triggered by microbial infections and is associated with severe systemic inflammatory response syndrome (SIRS).
- In addition to microbes, SIRS may be triggered by a variety of insults, including burns, trauma, and/or pancreatitis.

Less common types of shock:

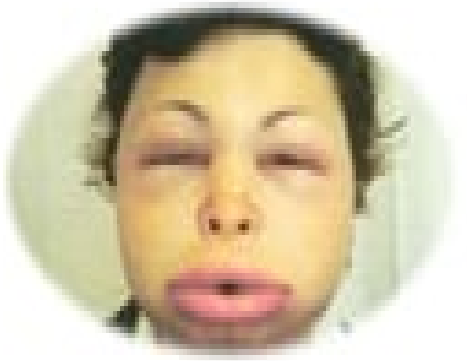
4. **Neurogenic shock:**

result from a loss of vascular tone associated with anesthesia or secondary to a spinal cord injury



5.Anaphylactic shock:

results from systemic vasodilation and increased vascular permeability that is triggered by an immunoglobulin E–mediated hypersensitivity reaction





Pathogenesis of Septic Shock:

- Despite improvements in care, the mortality of septic shock remains at a staggering 20% to 30%
- Septic shock is most frequently triggered by gram positive bacterial infections, followed by gram-negative bacteria and fungi. Hence, an older synonym, “endotoxic shock,” is no longer appropriate.

Factors believed to play major roles in the pathophysiology of septic shock include the following:

1. **Inflammatory and counterinflammatory responses:**

- various microbial cell wall constituents engage receptors on cells of the innate immune system, triggering proinflammatory responses.
- On activation, innate immune cells produce numerous cytokines, including TNF, IL-1, IFNγ, IL-12, and IL-18.

- Markers of acute inflammation such as C-reactive protein and procalcitonin are also elevated.
- The latter is a clinically useful indicator of septic shock.

- The hyperinflammatory state, initiated by sepsis, triggers counterregulatory immunosuppressive mechanisms, which may involve both innate and adaptive immune cells.

- As a result, septic patients may oscillate between hyperinflammatory and immunosuppressed states during their clinical course.
- Proposed mechanisms for the immune suppression include:

a) a shift from proinflammatory (TH1) to anti-inflammatory (TH2) cytokines
b) production of anti-inflammatory mediators (e.g., soluble TNF receptor, IL-1 receptor antagonist, and IL-10), lymphocyte apoptosis
c) the immunosuppressive effects of apoptotic cells
d) and the induction of cellular anergy

2. Endothelial activation and injury:

The proinflammatory state and endothelial cell activation associated with sepsis lead to widespread vascular leakage and tissue edema.

3. **Induction of a procoagulant state:**

The derangement in coagulation is sufficient to produce the formidable complication of disseminated intravascular coagulation in up to half of septic patients:

- Proinflammatory cytokines increase tissue factor production by monocytes, and decrease the production of endothelial anti-coagulant factors, such as tissue factor pathway inhibitor, thrombomodulin, and protein C.
- They also dampen fibrinolysis by increasing plasminogen activator inhibitor-1 expression.

- these effects lead to systemic activation of thrombin and the deposition of fibrin-rich thrombi in small vessels, often throughout the body, further compromising tissue perfusion.

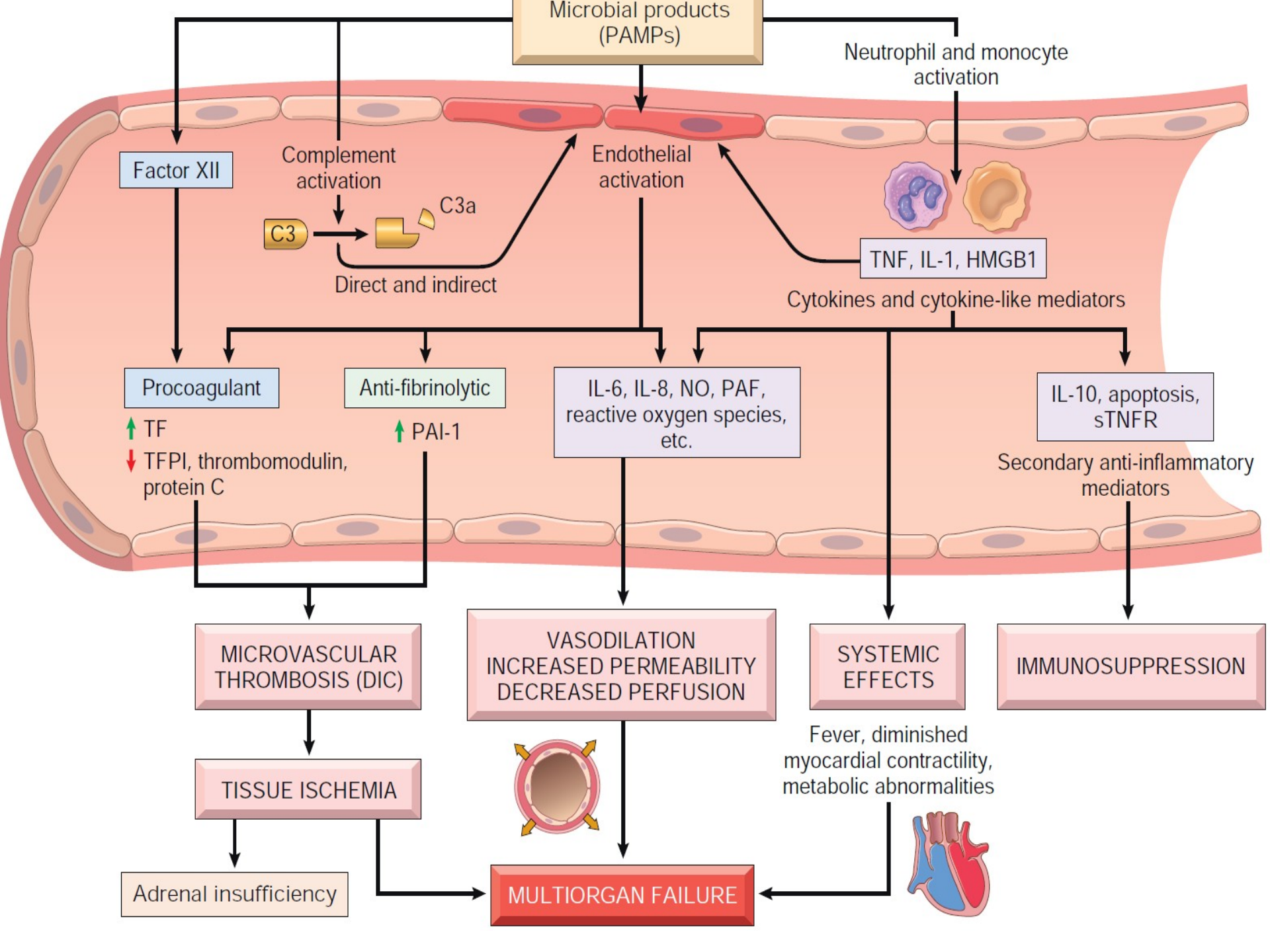
- In full-blown disseminated intravascular coagulation, the consumption of coagulation factors and platelets is so great that deficiencies of these factors appear, leading to concomitant bleeding and hemorrhage

4. **Metabolic abnormalities:**

- the proinflammatory cytokines suppress insulin release while simultaneously promoting insulin resistance
- Eventually hyperglycemia decreases neutrophil function—thereby suppressing bactericidal activity

5. **Organ dysfunction:**

- Systemic hypotension, interstitial edema, and small vessel thrombosis all decrease the delivery of oxygen and nutrients to the tissues (cellular hypoxia).
- Ultimately, these factors may conspire to cause the failure of multiple organs, particularly the kidneys, liver, lungs, and heart, culminating in death.


Microbial products (PAMPs)
Neutrophil and monocyte activation
Factor XII
Complement activation
C3
C3a
Direct and indirect
Endothelial activation
TNF, IL-1, HMGB1
Cytokines and cytokine-like mediators
Procoagulant
↑ TF
↓ TFPI, thrombomodulin, protein C
Anti-fibrinolytic
↑ PAI-1
IL-6, IL-8, NO, PAF, reactive oxygen species, etc.
IL-10, apoptosis, sTNFR
Secondary anti-inflammatory mediators
MICROVASCULAR THROMBOSIS (DIC)
VASODILATION
INCREASED PERMEABILITY
DECREASED PERFUSION
SYSTEMIC EFFECTS
IMMUNOSUPPRESSION
Fever, diminished myocardial contractility, metabolic abnormalities
TISSUE ISCHEMIA
Adrenal insufficiency
MULTIORGAN FAILURE

Stages of Shock:

Unless the insult is massive and rapidly lethal (e.g., a ruptured aortic aneurysm), shock tends to evolve through three general stages:

1. **An initial nonprogressive stage:** during which reflex compensatory mechanisms are activated and vital organ perfusion is maintained.
2. **A progressive stage:** characterized by tissue hypoperfusion and onset of worsening circulatory and metabolic derangement, including acidosis
3. **An irreversible stage:** in which cellular and tissue injury is so severe that even if the hemodynamic defects are corrected, survival is not possible.

Morphology:

- Although any organ can be affected, the brain, heart, kidneys, adrenals, and gastrointestinal tract are most commonly involved.
- Fibrin thrombi can form in any tissue but typically are most readily visualized in kidney glomeruli.
- Adrenal cortical cell lipid depletion reflects increased use of stored lipids for steroid synthesis
- In lung, diffuse alveolar damage leading to so-called “shock lung.”
- Except for neuronal and cardiomyocyte loss, affected tissues can recover completely if the patient survives.

Prognosis:

The severity and outcome of septic shock are likely dependent on:

1. the extent and virulence of the infection
2. the immune status of the host
3. the presence of other comorbid conditions
4. the pattern and level of mediator production.

Why in shock most attempts to intervene therapeutically with antagonists of specific mediators have not been effective and may even have had deleterious effects in some cases?

that is explained by the multiplicity of factors and the complexity of the interactions that underlie sepsis

- The standard of care remains antibiotics to treat the underlying infection and intravenous fluids, and supplemental oxygen to maintain blood pressure and limit tissue hypoxia.
- Suffice it to say that even in the best of clinical centers, septic shock remains an obstinate **clinical challenge.**

Summary

- Shock is defined as a state of systemic tissue hypoperfusion resulting from reduced cardiac output and/or reduced effective circulating blood volume.
- The major types of shock are cardiogenic (e.g., myocardial infarction), hypovolemic (e.g., blood loss), and septic (e.g., infections).
- Shock of any form can lead to hypoxic tissue injury if not corrected.
- Septic shock is caused by the host response to bacterial or fungal infections; it is characterized by endothelial cell activation, vasodilation, edema, disseminated intravascular coagulation, and metabolic derangements.